Tejas Kabra

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Available full-time for a Fall 2026 internship/co-op, August 2026 to January 2027

EDUCATION

Texas A&M University: B.S. in Electrical Engineering, GPA: 3.63 — Aug 2024 - May 2028

- Craig and Galen Brown Engineering Honors

EXPERIENCE

TEXAS A&M FSAE FORMULA ELECTRIC – LOW VOLTAGE ELECTRONICS TEAM — Aug 2025 - Present

- Designed a 4-layer STM32G4-based telemetry PCB integrating CAN, UART, and a SIM-enabled Quectel EG25 LTE modem to stream real-time vehicle diagnostics from the racecar to a Grafana-based driver station.
- Completed schematic capture and PCB layout across 2 board revisions, designing buck and LDO power stages to convert unregulated vehicle power for the microcontroller, LTE modem, and digital circuitry.
- Developed embedded firmware in C using STM32CubeIDE and HAL libraries to acquire CAN bus data, packetize 8-byte telemetry payloads at approximately 3200 packets per second, and transmit live vehicle metrics with approximately 20-30 ms of end-to-end latency.
- Validated board functionality using oscilloscopes and digital multimeters, verifying power rails, CAN/UART communication, and subsystem integration before deployment on the competition vehicle.

SOUTHLAND INDUSTRIES – ELECTRICAL DESIGN INTERN — May 2026 - July 2026

- Produced Revit electrical models and design documentation for modular cooling skids serving data center infrastructure, including power distribution, equipment connections, and coordinated electrical layouts.
- Performed power studies in SKM Power Tools and lighting simulations in AGI, applying NEC requirements and project design standards to verify distribution and lighting performance.
- Developed controls diagrams in Revit for modular cooling skids, documenting BACnet control architecture and coordinating electrical and controls interfaces.

PROJECTS

TEXAS A&M ROBOMASTERS – ELECTRICAL TEAM — Aug 2025 - Present

- Designed a custom STM32H7-based development board for robot subsystem control, integrating onboard power regulation, communication buses, sensor input, external programming access, and expanded GPIO.
- Implemented 24V-to-5V buck and 5V-to-3.3V LDO power stages while integrating an LSM6DSO32R IMU, crystal oscillator, CAN/UART connectivity, SWD access, and multifunction GPIO expansion.
- Created custom Altium component footprints, contributed to CAN breakout board development, and performed hands-on wiring, soldering, testing, STM32 programming, and subsystem integration with mechanical and software teams.

HIGH-SIDE TRANSIENT PROTECTION & LINEAR REGULATION CIRCUIT — Dec 2025 - Feb 2026

- Designed and simulated a high-side NMOS surge protection and linear regulation circuit for a 12 V, 5 A load, clamping the output during 50 V, 50 ms input transients.
- Designed the circuit around back-to-back N-channel MOSFETs, op-amp feedback, an LT1054 charge-pump gate drive, a zener-protected control rail, LDO regulation, and a current-sense shunt.
- Validated MOSFET linear-region operation, gate-source drive limits, output regulation, and short-duration surge power dissipation in LTspice.

AUTOMATED ELECTRONIC INVENTORY SYSTEM – TAMU FORMULA ELECTRIC — Aug 2025 - Dec 2025

- Built a Raspberry Pi-based automated inventory system using Python, I2C-controlled solenoid driver boards, and electromagnetic solenoid actuators to manage access to Formula Electric electrical components.
- Developed a touchscreen interface with card-based authentication and a controlled checkout workflow, allowing team members to access organized parts without manually searching inventory bins.
- Integrated and debugged GPIO/I2C communication, solenoid actuation, power delivery, and driver board behavior to improve reliability across repeated mechanical operation.

SKILLS

Tools: Altium, KiCad, Revit, SKM Power Tools, AGI, LTspice, STM32CubeIDE, Git, AutoCAD, SolidWorks, MATLAB, Excel

Hardware & Embedded: PCB Design, Schematic Capture, PCB Layout, Circuit Design, Oscilloscopes, Digital Multimeters, STM32, CAN, UART, I2C, BACnet, Soldering

Programming: C, C#, C++, Java, Python